

1.ArithmeticException

Q1 Handling ArithmeticException

10 marks

Write a Java program to divide two integers entered by the user. If the denominator is zero, handle the exception using a try-catch block and display an appropriate error message.

Your Code

```
1 import java.util.*;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int num1=sc.nextInt();
6         int num2=sc.nextInt();
7         try{
8             int res=num1/num2;
9             System.out.println("The result is : "+res);
10        }
11        catch(ArithmeticException e){
12            System.out.println("Number cant be divisible by zero.\nEnter any other number.");
13        }
14    }
15 }
```

Input

12
0

Output

Number cant be divisible by zero.
Enter any other number.

2. ArrayIndexOutOfBoundsException

Q2 Handling ArrayIndexOutOfBoundsException

10 marks

Write a Java program to create an integer array of five elements. Ask the user to enter an index and display the corresponding element. If the index is outside the valid range, handle the exception and display an error message.

Your Code

```
1 import java.util.*;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int[] a=new int[5];
6         System.out.print("Enter the elements in array:");
7         for(int i=0;i<5;i++){
8             a[i]=sc.nextInt();
9         }
10        try{
11            System.out.println("Enter an index:");
12            int n=sc.nextInt();
13            System.out.println("The element is: " + a[n]);
14        }
15        catch(ArrayIndexOutOfBoundsException e){
16            System.out.print("Invalid , enter the element within the array range.");
17        }
18    }
19 }
```

Input

8 9 6 7 3
6

Output

Enter the elements in array:Enter an index:
Invalid , enter the element within the array range.

3. Handling NumberFormatException

Q3 Handling NumberFormatException

10 marks

Write a Java program to accept a number as a string and convert it into an integer using Integer.parseInt(). If the input is not a valid integer, handle the exception and display an appropriate message.

Your Code

```
1 import java.util.*;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         String str=sc.nextLine();
6         try{
7             int i=Integer.parseInt(str);
8             System.out.println("The converted integer is: "+i);
9         }
10        catch (NumberFormatException e){
11            System.out.println("Invalid .Enter valid integer.");
12        }
13    }
14 }
```

Input

pooji

Output

Invalid .Enter valid integer.

4. Multiple Exception

Your Code

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int num1=sc.nextInt();
6         int num2=sc.nextInt();
7
8         try{
9             int result=num1/num2;
10            int[] a=new int[1];
11            a[5]=result;
12            System.out.println(num1+"/"+num2+": "+a[0]);
13        }
14        catch(ArithmeticException e){
15            System.out.println("Number cant be divided by zero.");
16        }
17        catch(ArrayIndexOutOfBoundsException e){
18            System.out.println("Index is out of Range , Enter a valid I
19        }
20    }
21 }
```

Input

4
0

Output

Number cant be divided by zero.